

Ayush Raj

Third Year Undergraduate
Department of Aerospace Engineering

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Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2023-Present	B.Tech	Indian Institute of Technology Kanpur	6.6/10
2023	CBSE(XII)	Indian Convent School,Kota	94.2%
2021	CBSE(X)	Army Public School,Sri Ganganagar	96.4%

Scholastic Achievements

- Secured All India Rank **3946** in **JEE Advanced 2023** among 2.5 lakh shortlisted candidates.
- Secured All India Rank **4175** in **JEE Mains 2023** among 11 lakh candidates.

Research Experience

SURGE'25 IIT KANPUR | B2B Brand Equity | Prof. R.N. Sengupta (May'25-July'25)

Objective	• To identify key determinants of B2B brand equity through literature review,data analysis of some companies and test some mathematical models using ML .
Approach	• Studied some journals related to B2B Brand Equity and analysed the models used in them to establish relationship between brand awareness,loyalty,trust,quality and reputation. • Used structural equation modelling and multiple logistic regression to analyse B2B brand equity of 5 B2B companies.Conducted research for MROI to check effect of digital marketing on B2B BE. •Analysed Semantic Brand Score to measure brand importance and strength using textual data and Monte Carlo Simulation for Brand Equity Risk Optimization
Impact	• Provided methods to measure B2B Brand Equity using various constructs and ML models and emphasized importance of digital marketing using MROI. .

Key Projects

Acrobatic Plane | Summer Project Under Aeromodelling Club (SnT) (May'24-July'24)

- Designed and fabricated a high-maneuverability acrobatic RC plane, applying principles of **aerodynamics** and flight stability.
- Integrated **electronic** components including ESCs, brushless motors, servos, and LiPo battery systems for responsive control.
- Modeled and tested **airfoil** shapes and control surface configurations to optimize lift-to-drag ratio and stunt performance.

Financial Markets & Investment Banking | Finlatics| Online Project 🔄 (Dec'24-Feb'25)

- Created and actively managed a real-time simulated equity portfolio benchmarked to the **BSE 500 Index**, applying principles of portfolio construction, diversification with a Finscore of **81.16**.
- Conducted sector-wise and completed a Financial Market Research Project comprising sectoral analysis using Porter's 5 Forces, portfolio restructuring and rebalancing, and strategic fund positioning, achieving an overall project rating of **82.12**.
- Gained conceptual and practical understanding of private equity investment cycles, including deal sourcing, evaluation, and capital deployment.
- Successfully completed a **Private Equity** and **Investment Banking** Research Project covering fundamentals, types of investments, fund setup, and case studies, achieving an overall project rating of **91.21**.

Hydro Mechanics Of Finvision | Google Developer Group Project 🔄 (Dec'24-Mar'25)

- Built scalable data extraction pipelines using **Selenium, Requests, and BeautifulSoup** in Python to scrape financial data (e.g., option chains, candlestick patterns) from dynamic websites like **Yahoo Finance,Screener and NSE**, enabling real-time market analysis.
- Applied advanced financial models including **Black-Scholes, Binomial Option Pricing, Vasicek, Cox-Ingersoll-Ross (CIR), and Merton Models** to price derivatives, assess risk, and simulate stochastic processes (**GBM, ADF, Weiner processes**) for market forecasting.
- Implemented **ML & deep learning algorithms** (e.g., **LSTM, RNN**) to predict candlestick patterns and price movements in crypto assets (ETH), and explored hybrid quantum-classical models using **TensorFlow Quantum and PennyLane** for enhanced feature extraction in trading strategies.

Assistive Devices:White Paper | IITK Consulting Group | Department Of SnT 🔄 (April'25-June'25)

- Worked under **Dr. Anita Aggarwal (DST, NAMEDD)** to develop a national white paper on assistive devices (ADs), aiming to attract investment and shape innovation policy.
- Co-authored the white paper with strategic recommendations based on global best practices and comparative analysis across USA, Japan, Germany, and Denmark.
- Built a **global AD leaderboard** and mapped **India's top 10 states** leading in medtech and healthcare innovation for elderly and disabled care.
- Evaluated **7+ government schemes** and identified **20+ VC firms, accelerators, and startup studios** active in the assistive tech ecosystem.

Nuvé: AI-Driven Fashion Trend Forecasting | Course Project-DES646 | Prof. A.K.Behera 🔄 (Sep'25-Nov'25)

- Developed an **AI-driven visual analytics system** for fashion trend forecasting as part of **DES646: AI/ML for Designers** at IIT Kanpur.
- Performed **sentiment analysis using NLP** on user comments and combined engagement metrics (likes, comments) to compute a composite **trend score**.
- Designed an interactive **Streamlit + Plotly dashboard** enabling designers to explore clusters, sentiment distributions, and emerging fashion patterns in real time.

Course Project-DMS624: Computer-Aided Decision Systems | Prof. Deepu Philip

(Aug'25-Nov'25)

- **OrbitWear: End-to-End Merchandise Decision System**
 - Identified merchandise pain points faced by IIT Kanpur clubs and teams, and provided an **end-to-end solution** covering **design, vendor coordination, pricing, order management, and delivery**.
 - Conducted **market validation surveys** across 10+ clubs/cells to validate demand and refine the offering based on real user needs.
 - Managed **pricing, advance payments, vendor payouts, and profit tracking** under a low-capital, pay-after-advance model to minimize financial risk.
- **Solving Farmers' Loan Access: Digital Kisan ID-Based Credit Framework**
 - Analyzed systemic barriers in agricultural credit access using **policy reports (NABARD, RBI, PM-KISAN)** and ground-level workflows.
 - Proposed a **Digital Kisan ID system** integrating Aadhaar, land records, crop data, and credit bureaus to auto-fill 80% of loan forms.
 - Designed a **secure, consent-based architecture** enabling instant credit retrieval, faster loan approvals, and reduced dependency on intermediaries.

TaxSmart | Capstone Project | Product Club IITK

(Dec'25)

- Conceptualized a **proactive tax planning product** addressing fragmented data, poor tax awareness, and last-minute filing behavior among Indian salaried and gig workers.
- Designed a structured **MVP feature set** including a Tax Planner Quiz, Personalized Tax Plan, Document Vault, Year-round Tax Projection, and a gamified **Tax Health Score**.
- Performed **competitive analysis** against Groww, ClearTax, and ET Money, identifying gaps in personalized planning, deduction tracking, and investment-linked tax guidance.
- Proposed a **product-led go-to-market strategy** with content-led onboarding, reminder-based retention loops, and monetization via insurance, ELSS, and NPS cross-sell.

Technical Skills

- **Programming Languages:** C,C++,Python,SQL
- **Software :** Canva,Arduino,MS Excel,Jupyter,Fusion 360
- **Libraries:** Numpy,Pandas,Tensorflow,Seaborn,Matplotlib,Keras,Scikit-Learn

Relevant Courses

Introduction To Management	Fundamentals of Computing - I	Application of Single Variable Calculus
Complex Variable	Linear Algebra	Ordinary Differential Equations
Single Variable Calculus	Partial Differential Equations	Fundamentals of Computing - II
Python for AI/ML/DL(online course)	Introduction To Electronics	Manufacturing Processes II
Computer-Aided Decision System	AI/ML for Designers	pi

Positions of Responsibility

Business Head | IITK Rocketry & Space Exploration Team

Mar'24-Present

- Worked in all technical subsystems like **avionics,structures,propulsion and recovery** .Helped the team in clearing various rounds of rocketry competitions. In July 2024, we launched our first rocket which soared to an apogee of **600ft**
- Oversaw financial documentation for funding proposals submitted to corporate sponsors and internal institute bodies; secured **2 sponsorship** .Managed the budget of around **5 lacks** and helped in clearance of all bills.
- Tested and calibrated load cells for one of India's highest-thrust **SRAD rocket motor (K95 %, 2500 N-s) under Project June**.Assisted in static fire testing and data acquisition setup for **Mission Prometheus's** K95-class solid rocket motor.
- Recruited and currently leading a team of **30+ juniors**; overseeing marketing, outreach, and sponsorship acquisition for RASET initiatives.Also participating as Recovery member in **INSPLACE Model Rocketry Competition**.

Marketing Secretary | Udghosh'24

Aug'24-Oct'24

- Gained hands-on experience in lead generation, **database management** of more than **100 companies**, and data analytics to support sponsorship outreach and marketing campaigns .
- Developed strong professional **communication skills**, including cold calling, formal email writing, and persuasive negotiation tailored to brand positioning.

Hospitality Secretary | Inter IIT Sports Meet'24

Nov'24-Dec'24

- Successfully managed hospitality operations for over **5,000 players** and officials from multiple IITs, ensuring smooth accommodation, meals, and logistics throughout the event.
- Utilized **Microsoft Excel** for room allocation, food distribution tracking, and schedule planning, increasing operational efficiency and reducing manual errors.

Extra-Curricular Activities

- Successfully completed the National Cadet Corps (NCC) '**C**' **Certificate** program, demonstrating commitment, leadership, discipline, and physical endurance.
- Represented Team RaSET in **IITK Samanvay: Industry Engagement Event 2025**, focusing on industry-academia collaboration. Showcased rocketry models and ongoing student-led aerospace projects to visiting industry experts, faculty, and students.
- Contributed to **Antarix'25 – National Space Day Celebration** Event (as part of Team RaSET). Conducted live exhibitions and explained rocketry concepts to school students, visitors, and participants to promote awareness of space science.
- Participating and cleared **PDR and CDR** round of **INSPEACE Model Rocketry Competition** as a part of IITK RaSET.
- Received **2 LORs and certificates** for outstanding performance in **finlatics financial markets investment banking** project.
- Received **certificate** for participating in **IIT Kanpur certification program on Python For AI,ML and DL**.